UNIVERSIDAD INTERCULTURAL PARA LA IGUALDAD

Programación e Innovación Digital

Materia: Base de Datos

PROYECTO INTEGRADOR: ABRAZA UN ARBOL

Presenta:

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Docente:

Mtra. María del Carmen Castañeda

Fecha de entrega:

24 de octubre de 2026

https://github.com/nohemiibarra1989z-lang/PROYECTO_ABRAZA_UN_ARBOL_DB.git

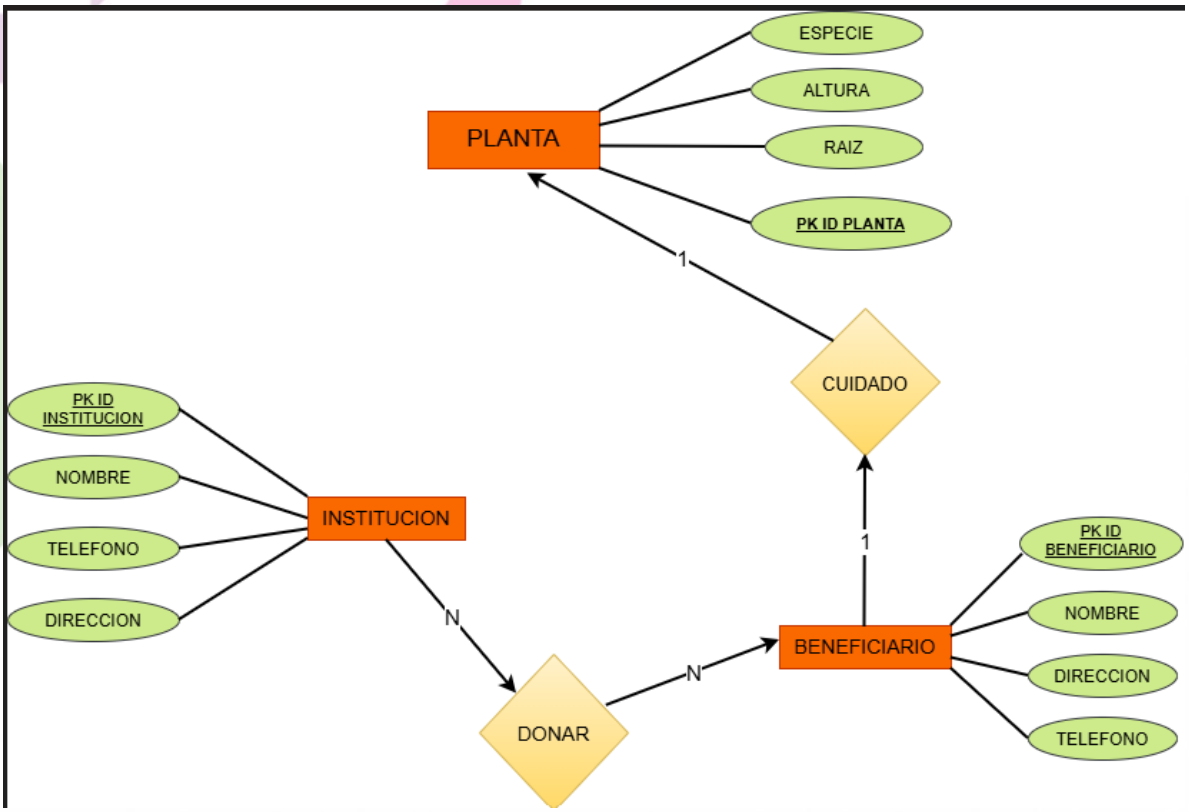


TABLA PLANTA

CAMPO	TIPO DE DATO	ROL	EXPLICACIÒN
ID_PLANTA	INT	PK	NUMERO UNICO PARA CADA PLANTA
ESPECIE	VARCHAR(100)	ATRIBUTO	NOMBRE PLANTA
ALTURA	VARCHAR(100)	ATRIBUTO	CUANTOS METROS CRECE
RAIZ	VARCHAR(100)	ATRIBUTO	TIPO DE RAIZ

TABLA BENEFICIARIO

CAMPO	TIPO DE DATO	ROL	EXPLICACIÒN
ID_BENEFICIARIO	INT	PK	IDENTIFICACIÒN OFICIAL
NOMBRE	VARCHAR(50)	ATRIBUTO	IDENTIFICACIÒN
DIRECCIÒN	VARCHAR(100)	ATRIBUTO	UBICACIÒN FISICA
TELÈFONO	VARCHAR(15)	ATRIBUTO	CONTACTO

TABLA INSTITUCIÒN

CAMPO	TIPO DE DATO	ROL	EXPLICACIÒN
ID_INSTITUCION	INT	PK	IDENTIFICACIÒN OFICIAL
NOMBRE	VARCHAR(50)	ATRIBUTO	IDENTIFICACIÒN
DIRECCIÒN	VARCHAR(100)	ATRIBUTO	UBICACIÒN
TELÈFONO	VARCHAR(15)	ATRIBUTO	CONTACTO

RELACIÒN 1:N

CAMPO	TIPO DE DATO	ROL	EXPLICACIÒN
ID_INSTITUCION	INT	PK	IDENTIFICACIÒN OFICIAL
NOMBRE_INSTITUCION	VARCHAR(100)	ATRIBUTO	TEXTO CORTO
DIRECCIÒN	VARCHAR(100)	ATRIBUTO	UBICACIÒN
TELEFONO	VARCHAR(15)	ATRIBUTO	CONTACTO
ID_PLANTA	INT	FK	NUMERO UNICO PARA CADA PLANTA

TABLA PLANA INICIAL (SIN NORMALIZAR)

ID_PLANTA	ESPECIE	ALTURA	RAIZ	ID_BENEFICIARIO	NOMBRE _BENEFICIARIO	DIRECCIÒN _BENEFICIARIO	TELÈFONO	ID_INSTITUCION	NOMBRE _INSTITUCIÒN	DIRECCIÒN_ INSTITUCIÒN	TELÈFONO
10	TRUENO	3 a 10metros	PIVOTANTE	443	ALEJANDRA HERRERA	GARAMBULLO #156	4492561305	22	VIVERO MUNICIPAL	JARDINES DEL PARQUE	4499723010
5	TRUENO	3 a 10metros	PIVOTANTE	654	MARISOL DE LEÒN	PITAHAYA #210	4494281620	46	VIVERO LA PALMA	SOLIDARIDAD 1	4492739258
1	TRUENO	3 a 10metros	PIVOTANTE	123	ANDREA MACIAS	SAGUARO #135	4498034961	55	VIVERO ESTATAL	ORTEGA DOUGLAS	4491020976

PRIMERA FORMA NORMAL (1FN)

ID_PLANTA	ESPECIE	ALTURA	RAIZ	ID_BENEFICIARIO	NOMBRE _BENEFICIARIO	DIRECCIÒN _BENEFICIARIO	TELÈFONO
10	TRUENO	3 a 10metros	PIVOTANTE	443	ALEJANDRA HERRERA	GARAMBULLO #156	4492561305
5	TRUENO	3 a 10metros	PIVOTANTE	654	MARISOL DE LEÒN	PITAHAYA #210	4494281620
1	TRUENO	3 a 10metros	PIVOTANTE	123	ANDREA MACIAS	SAGUARO #135	4498034961

SEGUNDA FORMA NORMAL (2FN)

PLANTA

ID_PLANTA (PK)	ESPECIE	ALTURA	RAIZ	ID_INSTITUCION (FK)
10	TRUENO	3 a 10metros	PIVOTANTE	22
5	TRUENO	3 a 10metros	PIVOTANTE	46
1	TRUENO	3 a 10metros	PIVOTANTE	55

INSTITUCIÒN

ID_INSTITUCION (PK)	NOMBRE_INSTITUCIÒN
22	VIVERO MUNICIPAL
46	VIVERO LA PALMA
55	VIVERO ESTATAL

BENEFICIARIO

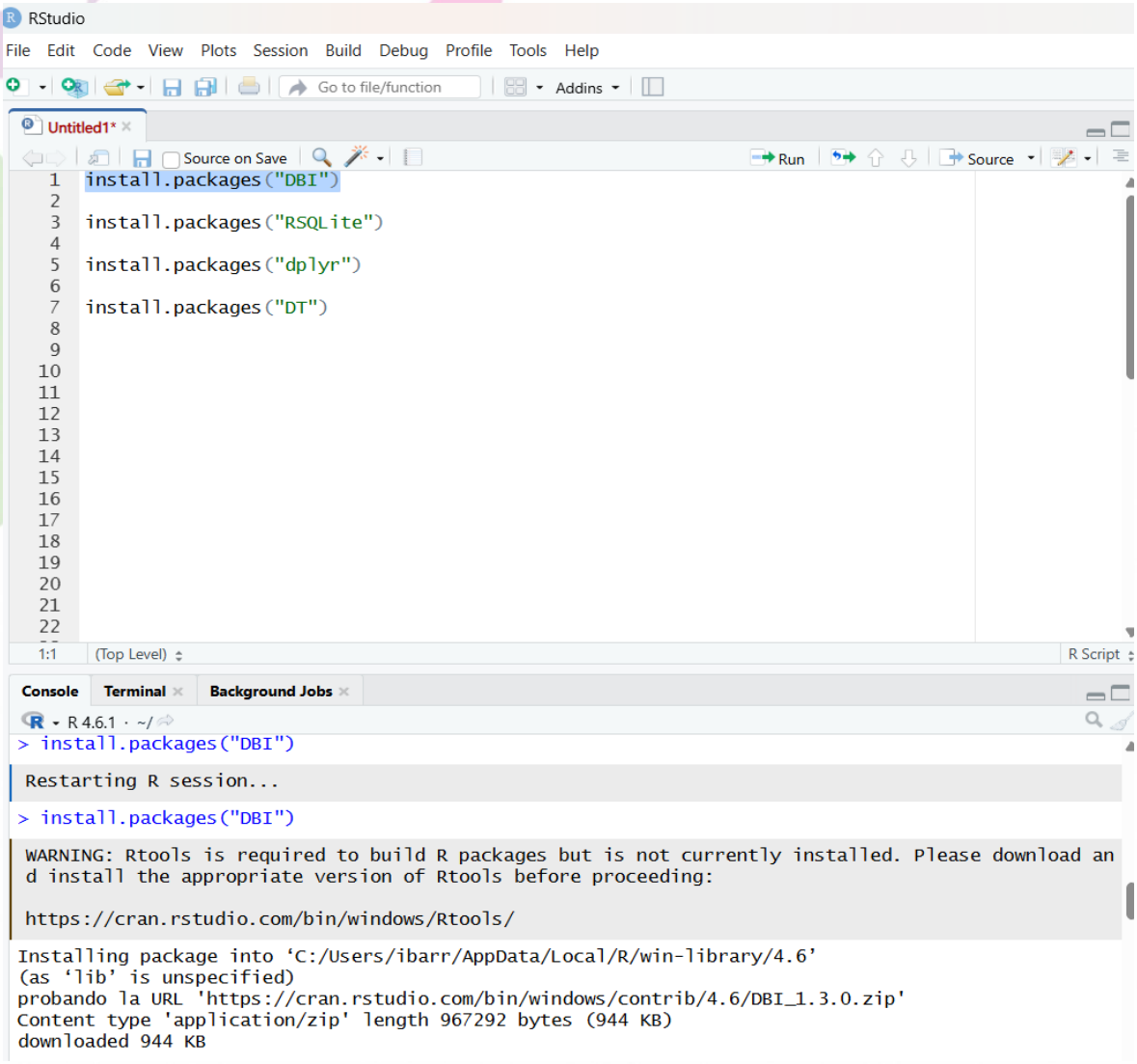
ID_BENEFICIARIO (PK)	TELÈFONO_ CONTACTO	ID_PLANTA (FK)
443	4492561305	10
654	4494281620	5
123	4498034961	1

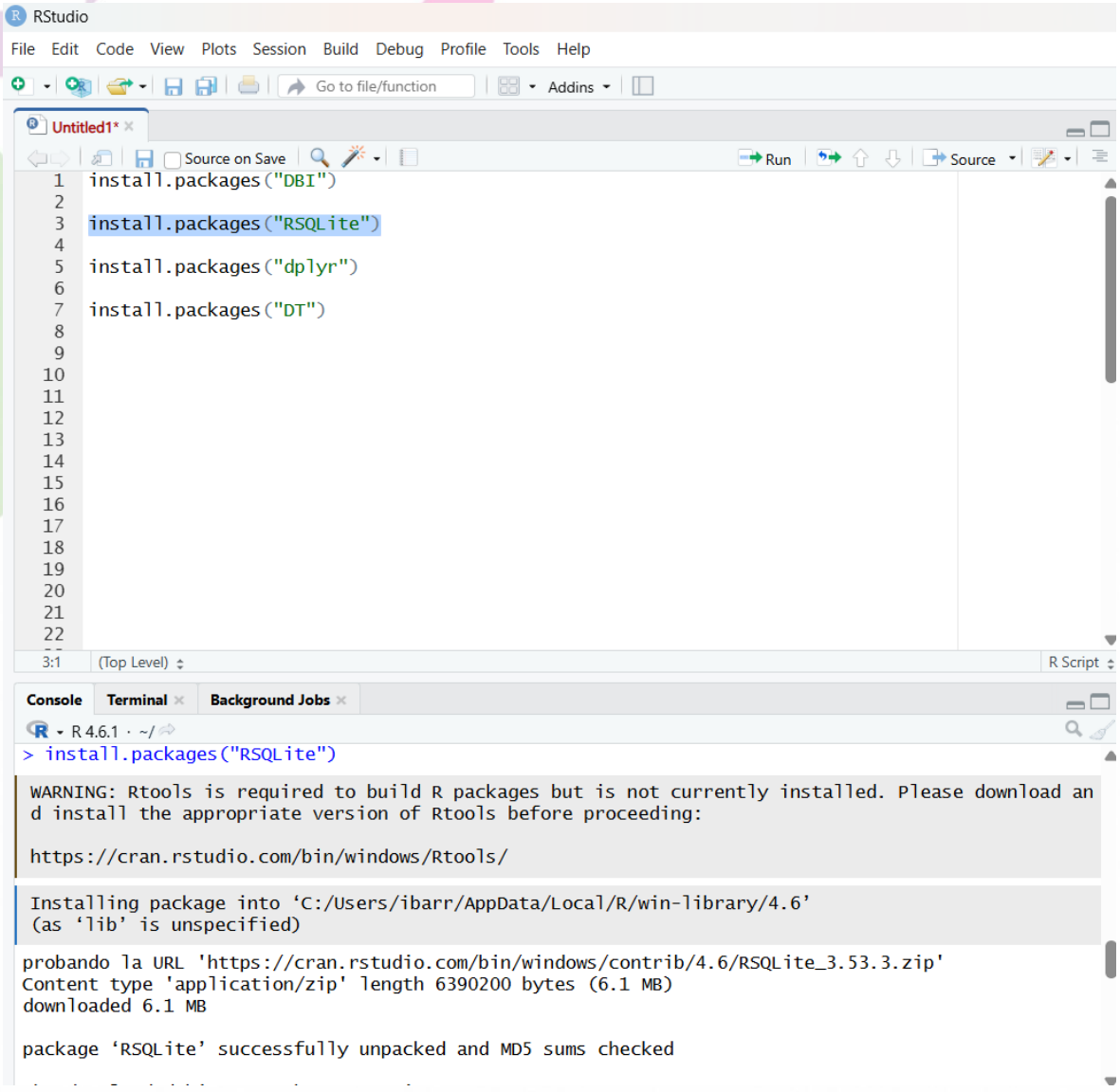
TERCERA FORMA NORMAL (3FN)

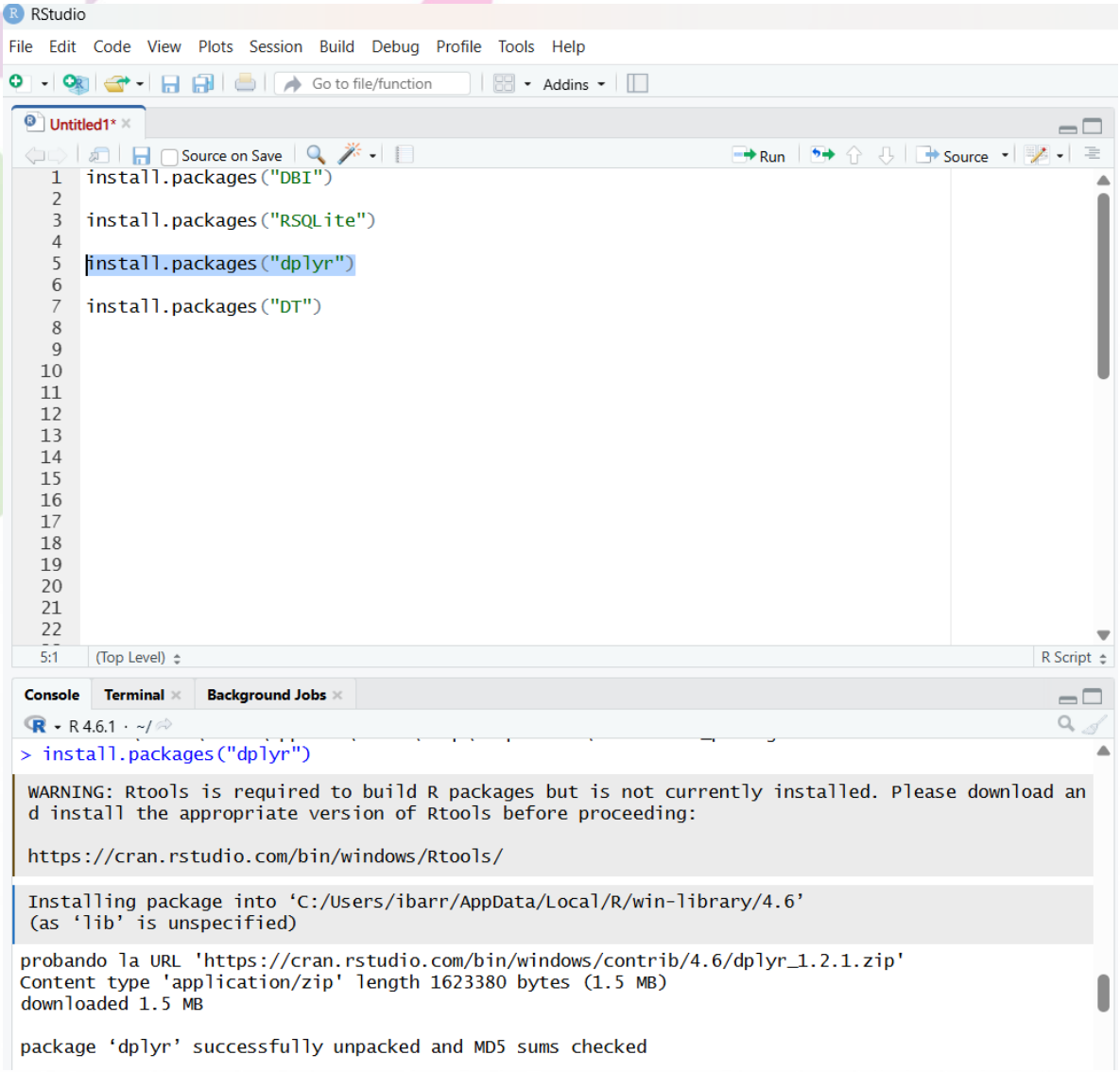
PLANTA		
PK	ID_PLANTA	INT
	ESPECIE	VARCHAR(100)
	ALTURA	VARCHAR(100)
	RAIZ	VARCHAR(100)
FK	ID_INSTITUCION	INT

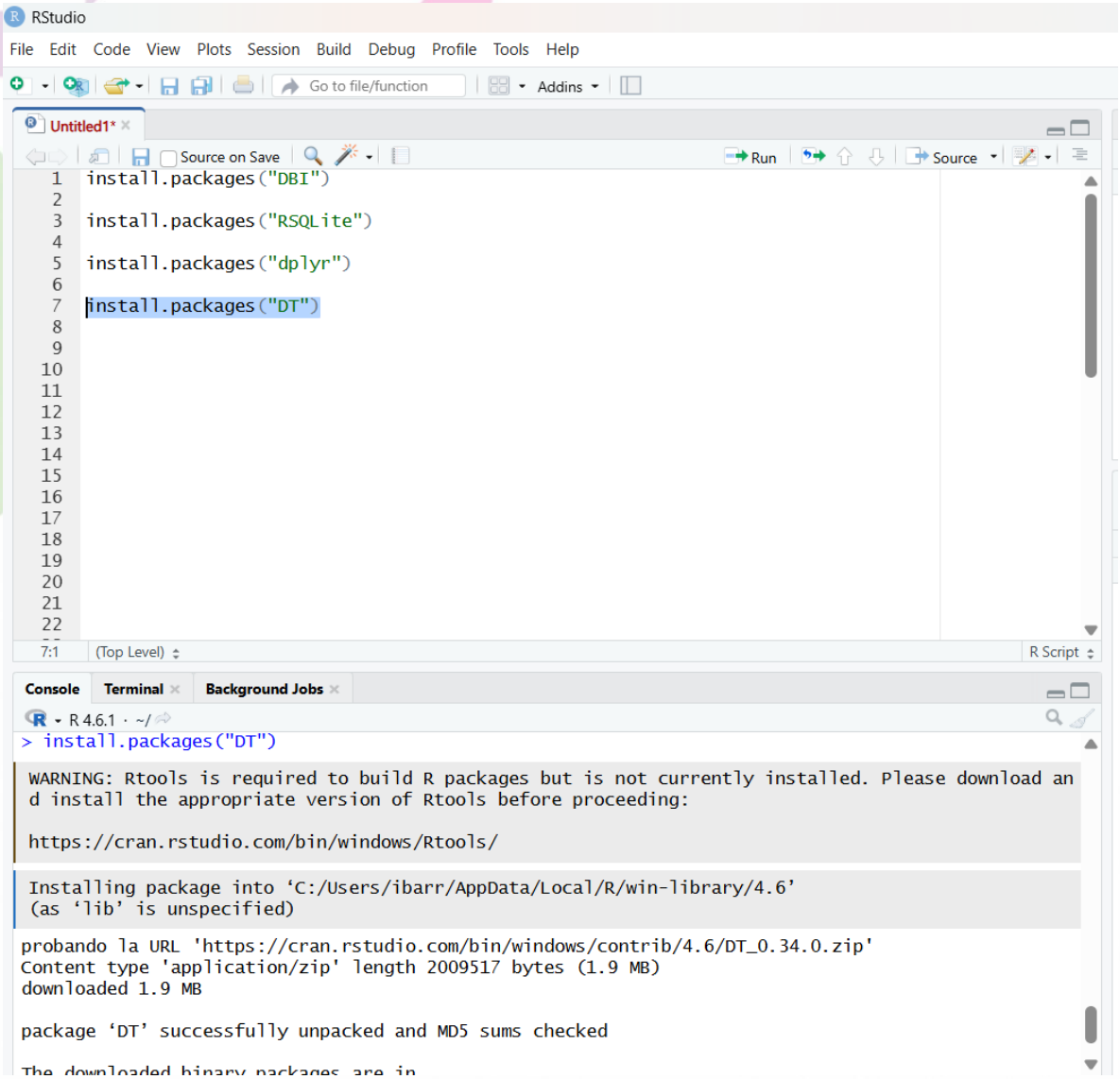
BENEFICIARIO		
PK	ID_BENEFICIARIO	INT
	NOMBRE	VARCHAR(50)
	DIRECCIÒN	VARCHAR(100)
	TELÈFONO	VARCHAR(15)
FK	ID_PLANTA	INT

INSTITUCIÒN		
PK	ID_INSTITUCION	INT
	NOMBRE	VARCHAR(50)
	DIRECCIÒN	VARCHAR(100)
	TELÈFONO	VARCHAR(15)
FK	ID_BENEFICIARIO	INT


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
1:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("DBI")
Restarting R session...
> install.packages("DBI")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/DBI_1.3.0.zip'
Content type 'application/zip' length 967292 bytes (944 KB)
downloaded 944 KB


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
3:1 (Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("RSQLite")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/RSQLite_3.53.3.zip'
Content type 'application/zip' length 6390200 bytes (6.1 MB)
downloaded 6.1 MB
package 'RSQLite' successfully unpacked and MD5 sums checked


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
5:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("dplyr")
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/dplyr_1.2.1.zip'
Content type 'application/zip' length 1623380 bytes (1.5 MB)
downloaded 1.5 MB
package 'dplyr' successfully unpacked and MD5 sums checked


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
Untitled1*
Source on Save
Run
Source
1 install.packages("DBI")
3 install.packages("RSQLite")
5 install.packages("dplyr")
7 install.packages("DT")
7:1
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> install.packages("DT")
WARNING: Rtools is required to build R packages but is not currently installed. Please download an
d install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Installing package into 'C:/Users/ibarr/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
probando la URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/DT_0.34.0.zip'
Content type 'application/zip' length 2009517 bytes (1.9 MB)
downloaded 1.9 MB
package 'DT' successfully unpacked and MD5 sums checked
The downloaded binary packages are in